

Chapter 2

Data Mining Procedures?:

- 1) *Classification* and *class probability estimation* attempt to predict, for each individual in a population, which of a (small) set of classes this individual belongs to (usually classes are mutually exclusive). Model produced should determine which class an individual belongs to when given a new individual. A closely related task is *scoring* or *class probability estimation*. A scoring model represents the probability (score) that an individual will fit into a class. Classification and scoring are models that can usually be modified to do the other.
- 2) *Regression* ("value estimation") attempts to estimate or predict, for each individual, the numerical value of some variable for that individual. Regression is related to classification, but the two are different. Classification predicts *whether* something will happen, whereas regression predicts *how much* something will happen.
- 3) *Similarity Matching* attempts to identify similar individuals based on data known about them. Similarity measures underlie certain solutions to other data mining tasks, such as classification, regression, and clustering.
- 4) *Clustering* attempts to group individuals in a population together by their similarity, but not driven by any specific purpose. Useful in preliminary domain exploration to see which natural groups exist because these groups in turn may suggest other data mining tasks or approaches.
- 5) *Co-occurrence grouping* (aka frequent itemset mining, association rule discovery, and market-basket analysis) attempts to find associations between entities based on transactions involving them. This grouping considers similarity of objects based on their appearing together in transactions.
- 6) *Profiling* (aka behavior description) attempts to characterize the typical behavior of an individual, group, or population. Often used to establish behavioral norms.
- 7) *Link Prediction* attempts to predict connections between data items, usually by suggesting that a link should exist, and possibly also estimating the strength of the link. Common in social networking systems.
- 8) *Data Reduction* attempts to take a large set of data and replace it with a smaller set of data that contains much of the important information in the larger set. Usually involves loss of information, but what is important is the trade-off for improved insight.
- 9) *Casual Modeling* attempts to help us understand what events or actions actually influence others, i.e. A/B test. Look out for placebo effects!

Supervised vs Unsupervised:

Unsupervised – When there is no such target the data mining problem is thought to be unsupervised.

Supervised – When a specific target is defined this is supervised.

*Vital part of early stages of data mining process is to decide whether the line of attack will be supervised/unsupervised, and if supervised, to produce a precise definition of a target variable.

Page. 27 for the CRISP model (write in from tablet).

Purity Measure is formula based on that evaluates how well each attribute splits a set of examples into segments, with respect to a chosen target variable. The most common splitting criterion is called *information gain* and based on purity measure called *entropy*. Entropy is a measure of disorder that can be applied to a set.

$$\text{Entropy} = -p_1 \log(p_1) - p_2 \log(p_2) - \dots$$

